Non-stationary Time Series

Operators · Unit roots · ADF / KPSS · ARIMA / SARIMA

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Learning objectives

By the end of this chapter, you will be able to:

- recognise the limits of the stationarity assumption and the danger of *spurious regressions*;
- distinguish *trend stationary* (TS) from *difference stationary* (DS, with a unit root);
- manipulate the *backshift* B and *difference* $\Delta = 1 - B$ operators;
- run the *ADF* and *KPSS* unit-root tests and read their conclusions correctly;
- formulate *ARIMA(p, d, q)* and *SARIMA(p, d, q)(P, D, Q)$_s$* models.

Chapter outline

1. Why non-stationarity matters (spurious regression)
2. Backshift and difference operators
3. Trend-stationary vs difference-stationary
4. Unit-root tests: ADF, KPSS, Phillips–Perron
5. ARIMA(p, d, q)
6. Seasonal ARIMA: SARIMA(p, d, q)(P, D, Q)$_s$

Why non-stationarity matters

1. Limits of the stationarity assumption

Most real-world series — GDP, asset prices, populations, cumulative indices — are *not* stationary:

- their mean drifts upward / downward over time;
- their variance often grows;
- shocks can have permanent effects.

Spurious regression (Granger & Newbold, 1974)

Regressing one random walk on another *independent* random walk yields a high R^2 and a "significant" t-statistic, even though there is no causal link. This famous fallacy is avoided only by working with stationary (or co-integrated) series.

1. Two flavours of non-stationarity

Definition – Trend stationary (TS)

$X_t = f(t) + \varepsilon_t$ with ε_t stationary. Removing the deterministic trend $f(t)$ recovers a stationary process. Shocks have *transient* effects.

Definition – Difference stationary (DS)

$\Delta X_t = X_t - X_{t-1}$ is stationary, but X_t has a *unit root* (e.g. random walk). Shocks are *permanent*.

Same plot can be *either* TS or DS — visual inspection alone is not enough. Hence the need for a formal *unit-root test*.

Backshift and difference operators

2. Backshift operator B

Definition – Backshift

$BX_t = X_{t-1}$, $B^k X_t = X_{t-k}$.

Algebra of B. For any polynomials $\phi(z), \theta(z)$:

$$\phi(B)\,\theta(B) = \theta(B)\,\phi(B), \quad \text{and} \quad (1 - \phi B)X_t = X_t - \phi X_{t-1}.$$

Compact AR(1): $X_t = \phi X_{t-1} + \varepsilon_t \Leftrightarrow (1 - \phi B)X_t = \varepsilon_t$.

- Stationarity $\Leftrightarrow$ all roots of $\phi(z) = 0$ lie *outside* the unit disk.
- $\phi(z) = 1 - \phi z$ has a root at $z = 1/\phi$: stationary iff $|\phi| < 1$.

2. Difference operator Δ

Definition – Difference

$$\Delta X_t \; = \; X_t - X_{t-1} \; = \; (1 - B)X_t.$$

Higher orders:

$$\Delta^2 X_t = \Delta(\Delta X_t) = X_t - 2X_{t-1} + X_{t-2} = (1 - B)^2 X_t.$$

Seasonal difference:

$$\Delta_s X_t = X_t - X_{t-s} = (1 - B^s)X_t.$$

A series is *integrated of order d*, written $X_t \sim I(d)$, if $\Delta^d X_t$ is stationary but $\Delta^{d-1} X_t$ is not.

2. Random walk: archetypal $I(1)$ process

$X_t = X_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$.

$$\mathrm{Var}[X_t] = t\,\sigma^2 \to \infty, \qquad \Delta X_t = X_t - X_{t-1} = \varepsilon_t \in \mathrm{WN}(0, \sigma^2).$$

With drift: $X_t = c + X_{t-1} + \varepsilon_t \Rightarrow \Delta X_t = c + \varepsilon_t$ — stationary mean c.

A random walk *does not* fluctuate around a fixed level. It wanders, with a variance that grows linearly in t. This is the null hypothesis of the ADF test.

Unit-root tests

3. Augmented Dickey–Fuller (ADF) test

Model: $X_t = \rho X_{t-1} + \varepsilon_t$. Subtracting X_{t-1}:

$$\Delta X_t \;=\; \gamma\, X_{t-1} + \varepsilon_t, \qquad \gamma = \rho - 1.$$

Augmented version (adds lags of ΔX_t to whiten residuals):

$$\Delta X_t \;=\; \gamma\, X_{t-1} + \sum_{i=1}^{p} \alpha_i\, \Delta X_{t-i} + \varepsilon_t.$$

Definition – Hypotheses

H_0: $\gamma = 0$ — **unit root, non-stationary**.

H_1: $\gamma < 0$ — stationary (mean-reverting).

Reject H_0 if the t-statistic for $\widehat{\gamma}$ is more negative than the Dickey–Fuller critical values (~ -2.86 at 5% with intercept; *not* the standard normal!).

3. ADF: practical decision rule and traps

- Use the *augmented* version with p chosen by AIC/BIC.
- Three flavours: no constant, with constant, with constant and trend. Choose based on the visual pattern of the series.
- Power is low when ρ is close to 1 (near unit root).

Be careful with the *direction of the conclusion*:

- p-value $< 0.05 \Rightarrow$ **reject** $H_0 \Rightarrow$ stationary;
- p-value $> 0.05 \Rightarrow$ *do not reject* $H_0 \Rightarrow$ non-stationarity is *not ruled out* (this is *not* "the series is stationary").

3. KPSS test (opposite null hypothesis)

Definition – KPSS — Kwiatkowski et al., 1992

H_0: **stationarity** (around a level or a trend).

H_1: unit-root non-stationarity.

Reject H_0 at 5% if the test statistic exceeds 0.463 (level) or 0.146 (trend) — depending on the version.

Practical use: ADF + KPSS together

- ADF rejects + KPSS does not reject $\Rightarrow$ stationary.
- ADF does not reject + KPSS rejects $\Rightarrow$ non-stationary.
- ADF rejects + KPSS rejects $\Rightarrow$ inconclusive (probably structural break).
- ADF does not reject + KPSS does not reject $\Rightarrow$ insufficient information (small sample).

3. ADF and KPSS in Python

```
from statsmodels.tsa.stattools import adfuller, kpss

# Augmented Dickey-Fuller
adf_stat, adf_p, adf_lag, adf_n, adf_crit, _ = adfuller(y)
print(f"ADF stat={adf_stat:.3f}, p-value={adf_p:.4f}")

# KPSS (level by default)
kpss_stat, kpss_p, kpss_lag, kpss_crit = kpss(y, regression="c")
print(f"KPSS stat={kpss_stat:.3f}, p-value={kpss_p:.4f}")
```

ARIMA

4. ARIMA(p, d, q)

Definition – ARIMA(p, d, q)

A process X_t is *ARIMA(p, d, q)* if its d-th difference is a stationary ARMA(p, q) :

$$\phi(B)\,(1-B)^d\,X_t \;=\; \theta(B)\,\varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0,\sigma^2),$$

with $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$.

- Bring X_t to stationarity by differencing d times.
- Fit ARMA(p, q) on the differenced series.
- Forecasts on the original scale are obtained by integrating back ($X_t = X_{t-1} + \Delta X_t$).

4. Box–Jenkins methodology

1. **Plot** X_t. Identify trend, seasonality, level shifts.
2. **Test** for unit roots (ADF + KPSS). If non-stationary, difference until stationary $\Rightarrow$ choose d.
3. **Identify** (p, q) from the ACF/PACF of the differenced series (Box–Jenkins signature, Chapter 3).
4. **Estimate** the ARMA(p, q) coefficients (MLE).
5. **Diagnose**: Ljung–Box on residuals, plot of residuals.
6. **Forecast** and integrate back.

4. Example: ARIMA(1, 1, 1)

Definition: $\phi(B)(1-B)X_t = \theta(B)\varepsilon_t$ with $\phi(B) = 1 - \phi B$ and $\theta(B) = 1 + \theta B$.
Expanded:

$$(1-\phi B)(1-B)X_t = (1+\theta B)\varepsilon_t.$$

Letting $W_t = \Delta X_t$:

$$W_t = \phi W_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}.$$

W_t is an ARMA(1,1); X_t is its cumulative sum starting from X_0.

Seasonal ARIMA (SARIMA)

5. SARIMA(p, d, q)(P, D, Q)$_s$

Definition – SARIMA

$$\phi(B)\,\Phi(B^s)\,(1-B)^d\,(1-B^s)^D\,X_t \;=\; \theta(B)\,\Theta(B^s)\,\varepsilon_t,$$

with seasonal AR/MA polynomials Φ, Θ in B^s.

Six orders:

- **Non-seasonal**: p (AR), d (regular differencing), q (MA);
- **Seasonal**: P (SAR), D (seasonal differencing), Q (SMA);
- **Period**: s (e.g. 12 for monthly data).

5. Example: SARIMA(1, 1, 1)(1, 1, 1)$_{12}$

$$(1-\phi B)(1-\Phi B^{12})(1-B)(1-B^{12})X_t = (1+\theta B)(1+\Theta B^{12})\varepsilon_t.$$

- Differences once at lag 1 *and* once at lag 12.
- Captures both short-term persistence and yearly seasonality.
- Standard for monthly economic, climate and sales data.

5. SARIMA in Python

```python
from statsmodels.tsa.statespace.sarimax import SARIMAX

model = SARIMAX(y,
                order=(1, 1, 1),
                seasonal_order=(1, 1, 1, 12))
fit = model.fit(disp=False)
print(fit.summary())

# 12-month ahead forecast
forecast = fit.get_forecast(steps=12)
mean = forecast.predicted_mean
ci = forecast.conf_int(alpha=0.05)
```

Up next

Chapter 5. ARMA models — estimation, diagnostics, forecasting in detail.

- AR(p), MA(q), ARMA(p, q) algebra.
- Yule–Walker, MLE estimation.
- Order selection (AIC, BIC).
- Forecasting and prediction intervals.

Thank you.

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